

Description & Eligibility

Contestants will demonstrate foundational knowledge of network administration within a Cisco® environment. This event focuses on essential skills in configuring, managing, and securing network devices and architectures. Any secondary or postsecondary division contestant may enter this objective event.

Contestant Must Supply

- Published and/or unpublished non-electronic written reference materials, including hands-on tasks

Equipment/Supplies Provided

- Computer and software

Contest Notes

- Method of evaluation: objective
- AI Policy Level 1 *No AI Allowed*
- Previous Business Professionals of America tests and/or sample tests (handwritten, photocopied or keyed) may not be taken into the testing area. Violation of this rule will result in disqualification.
- At NLC, contestants will take Cisco Certified Support Technician certification as part of the event. Contestants passing this certification will have 50 points added to their event score.
 - A Pearson profile must be created at <https://certiport.pearsonvue.com> before attending NLC
 - Contestants must include their BPA Contestant ID in their Pearson profile when they register
 - Certiport Parent/Legal Guardian Consent Form must be uploaded to: education@bpa.org
 - For more information on the exam, visit <https://certiport.pearsonvue.com>
 - Reference materials are not allowed for certification testing
- The top 10 contestants at Nationals will advance to a final round that includes a performance component. Scores from the preliminary round and the objective test will not carry over to the finals. Reference materials may be used during the application portion of the final round.
- Length of event
 - Preliminaries - No more than sixty (60) minutes for objective test
 - Preliminaries (Nationals only) - No more than sixty (60) minutes for objective test, no more than fifty (50) minutes for certification test, and no more than sixty (60) minutes for performance component
 - Finals - (Nationals only) No more than sixty (60) minutes for performance component
- Entries vary by state

Contest Competencies

1. Design, configure, secure, and optimize routing, switching, VLANs, and First Hop Redundancy Protocols (FHRP) to ensure efficient, segmented, and highly available enterprise networks
2. Design and implement scalable LAN/WAN topologies using advanced routing, switching, and security protocols to support resilient and secure network infrastructures
3. Configure and manage NAT, IPv4/IPv6 addressing, and access control lists (ACLs) to secure and control network traffic in modern environments
4. Implement network management solutions to monitor, analyze, and maintain network health, performance, and security
5. Explain the fundamentals of Software-Defined Networking (SDN) and how centralized control, programmability, and abstraction improve network flexibility, automation, and security
6. Use network automation tools to streamline configuration, reduce errors, and improve efficiency in managing network devices and services
7. Configure wireless LAN controllers (WLCs) to manage access points, apply wireless policies, and ensure secure, scalable, and high-performance wireless network operations
8. Plan and deploy wireless access points by analyzing coverage needs, interference, capacity requirements, and best-practice placement to support reliable and efficient wireless connectivity